01 THE NUR DOME

EARTHBAG / SUPERADOBE

The heart of the land: a 29-ft earth dome, with corridors leading to domes 2 and 3.

LEVEL	CREW	WHERE	STAGE NOW
■■□ Some skill or supervision	3–6	Open ground between the sheds and the old foundation	Foundation trench + final plan

The Nur Dome is the 29-ft central dome: the prayer and meeting space. Two corridors will link it to **Dome 2 (Hisham)**, the smaller bedroom dome toward the sheds, and **Dome 3 (Nazim)**, the smallest, the meditation dome, angled toward the entrance. Right now the work is on the ground: settling the plan and digging the foundation trench. Every bag of earth you lay here is part of the building.

FIRST STEPS *today*

1. Find the marked center point. Set a sturdy center post there (steel pipe, driven deep or set in concrete). This is the compass pivot for the whole dome.
2. Scribe two circles with a string and lime or flour: the inside face of the wall (14 ft 6 in radius) and the outside face (inside radius + bag width, about 16–18 in more).
3. Mark three lines out from the circle: the main door (facing the koi pond), the corridor to Hisham and the corridor to Nazim.
4. Dig the trench between the circles, about 12–18 in deep, down to firm undisturbed soil. Keep the bottom level (water level or laser).
5. Sort the spoil: topsoil to the garden pile, subsoil to the bag pile. Fill two jars with subsoil and water for a jar test.

BEFORE YOU START

- Call 811 (DigAlert) before any digging, and ask Yusuf where the water lines from the shared tank run.
- Filled tubes are heavy. Lift with your legs, work in pairs, and swap jobs every hour.
- Barbed wire goes between every course. Wear leather gloves and eye protection when handling it.
- Lime and cement are caustic. Wear gloves, glasses and a dust mask when mixing; rinse skin right away.
- Shade, water and a break every hour. Earth work in the sun is harder than it looks.

THE BASICS

SuperAdobe	Long polypropylene tubes are filled with damp earth right on the wall, then tamped flat. Two strands of barbed wire between courses act like Velcro and hold the walls in tension.
The compass	A fixed center post with a measuring arm traces every course in a perfect circle. A second, height compass tells you how far each course steps inward, so the walls lean in and close into a dome.
The soil	Good bag earth has enough clay to bind but is mostly sand and grit. Lower courses are often stabilized with lime (traditional) or around 10% cement. A jar test tells you what you have.
Dry feet	The first courses sit in a rubble trench and are filled with gravel, not earth, so ground moisture can't wick up into the walls.

Size check	29 ft is larger than most earthbag domes built to date. The plan should be reviewed by an engineer experienced with earthbag before the walls rise above shoulder height.

THE PLAN *in order*

01	**Finalize the plan**	Dome profile, door direction, window heights, corridor lengths, skylight, stabilizer (lime or cement), engineer review and permit route.
02	**Foundation**	Trench, compacted gravel base, 2–3 courses of gravel-filled tube, barbed wire between each.
03	**Walls**	Earth courses traced with the compass. Arch forms placed for the door and windows at set heights. Buttresses where the corridors meet.
04	**Corridors**	Corridor trenches dug once the main ring is a few courses up, so they key into the dome rather than butt against it.
05	**Closing the dome**	Upper courses stepped in by the height compass; skylight or oculus framed at the crown.
06	**Plaster**	Earthen and lime plaster coats; local slate from the mountain.

TOOLS & MATERIALS

SuperAdobe tubing (18 in) · 4-point barbed wire · Tube feeding chute (sheet-metal slider) · Hand tampers · Buckets / coffee cans for filling · Center + height compass · Sifting screen · Shovels, mattock, wheelbarrows · Water level or laser level · Leather gloves, safety glasses, dust masks · Bricks to weigh down the wire · Tarps to keep the earth damp

FOR YUSUF TO DECIDE

- ? Pointed or catenary dome profile?
- ? Exact door bearing toward the koi pond
- ? Corridor lengths to Hisham and Nazim
- ? Lime or cement stabilizer
- ? Engineer + permit route
- ? Skylight / oculus at the crown

Resources: CalEarth Institute (SuperAdobe, Hesperia) (calearth.org) · Book: *Earthbag Building*, Kaki Hunter & Donald Kiffmeyer